

2.

(1). Stark 效应: 原子能级在外电场中发生分裂的现象
 Zeeman 效应: 原子能级在外磁场中发生分裂的现象

$$(2) \quad [\hat{L}_x, \hat{L}_y] = [y \frac{\partial}{\partial z} - z \frac{\partial}{\partial y}, z \frac{\partial}{\partial x} - x \frac{\partial}{\partial z}] \psi \quad (1)$$

$$= [y \frac{\partial}{\partial z}, z \frac{\partial}{\partial x}] \psi + [z \frac{\partial}{\partial x}, y \frac{\partial}{\partial z}] \psi - [z \frac{\partial}{\partial y}, z \frac{\partial}{\partial x}] \psi - [x \frac{\partial}{\partial z}, x \frac{\partial}{\partial z}] \psi$$

$$= (y \frac{\partial}{\partial x} [z \frac{\partial}{\partial z}, z] \psi - x \frac{\partial}{\partial y} [z \frac{\partial}{\partial z}, z] \psi)$$

$$= [y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y}] \psi$$

$$= (1\hbar) [y \frac{\partial}{\partial x} - x \frac{\partial}{\partial y}] \psi$$

$$= \hat{L}_z \psi$$

$$[\hat{L}_x, \hat{L}_y] = [y \hat{p}_z - z \hat{p}_y, z \hat{p}_x - x \hat{p}_z] \psi$$

$$= [y \hat{p}_z, z \hat{p}_x] - [y \hat{p}_z, x \hat{p}_z] - [z \hat{p}_y, z \hat{p}_x] + [z \hat{p}_y, x \hat{p}_z]$$

$$= -i\hbar y \hat{p}_x + i\hbar x \hat{p}_y$$

$$= i\hbar [x \hat{p}_y - y \hat{p}_x]$$

$$= \hbar \hat{L}_z$$

$$\text{同理有 } [\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x, [\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$$

$$[\hat{L}_x, \hat{L}_y] = i\hbar \hat{L}_z$$

(3). E_n^2 求解

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$$(4) \quad \frac{d\langle \hat{E} \rangle}{dt} = \langle \hat{H} | \hat{H} | \psi \rangle$$

$$\frac{d\langle \hat{E} \rangle}{dt} =$$

$$(4) \quad \frac{d\langle \hat{E} \rangle}{dt} = \frac{d}{dt} \langle \psi | \hat{H} | \psi \rangle$$

$$= \langle \frac{d\psi}{dt} | \hat{H} | \psi \rangle + \langle \psi | \frac{d\hat{H}}{dt} | \psi \rangle + \langle \psi | \hat{H} | \frac{d\psi}{dt} \rangle$$

$$= E \langle \frac{d\psi}{dt} | \psi \rangle + \langle \psi | \frac{d\hat{H}}{dt} | \psi \rangle + \langle \psi | \hat{H} | \frac{d\psi}{dt} \rangle$$

$$= E \frac{d\langle \psi | \psi \rangle}{dt} + \langle \psi | \frac{d\hat{H}}{dt} | \psi \rangle$$

$$= \langle \psi | \frac{d\hat{H}}{dt} | \psi \rangle$$

Feynman-Hellmann

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{d}{dt} \langle \psi | \hat{A} | \psi \rangle$$

$$= \langle \frac{d\psi}{dt} | \hat{A} | \psi \rangle + \langle \psi | \frac{d\hat{A}}{dt} | \psi \rangle + \langle \psi | \hat{A} | \frac{d\psi}{dt} \rangle$$



由 $i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle$

$\frac{\partial}{\partial t} |\psi\rangle = \frac{1}{i\hbar} \hat{H} |\psi\rangle$

$\langle \frac{\partial}{\partial t} \psi | = \frac{1}{i\hbar} \langle \psi | \hat{H}$

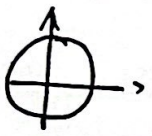
$\therefore \hat{H} = -\frac{1}{2\mu} \langle \psi | \hat{p}^2 | \psi \rangle + \langle \psi | \frac{\partial \hat{H}}{\partial t} | \psi \rangle + \frac{1}{i\hbar} \langle \psi | \hat{H} | \psi \rangle$

$= \frac{1}{2\mu} \langle \psi | \hat{p}^2 | \psi \rangle + \langle \psi | \frac{\partial \hat{H}}{\partial t} | \psi \rangle$

$= \frac{1}{2\mu} \langle \psi | [\hat{A}, \hat{H}] | \psi \rangle + \langle \psi | \frac{\partial \hat{H}}{\partial t} | \psi \rangle$

(5)
$$V(x) = \begin{cases} \frac{1}{2} m v^2, & |x| < a \\ \infty, & |x| > a \end{cases}$$

(6)



$\hat{H} = \frac{1}{2\mu} (\hat{p} - \frac{qA}{c})^2$

$B = \nabla \cdot \vec{A}$

选取规范 $\vec{A} = (-\frac{By}{2}, \frac{Bx}{2}, 0)$

则
$$\hat{H} = \frac{1}{2\mu} [(\hat{p}_x + \frac{qBy}{2c})^2 + (\hat{p}_y - \frac{qBx}{2c})^2 + \hat{p}_z^2]$$

(7) 由能量与动量守恒可得
自由粒子的能量与动量的关系为

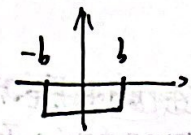
$\Delta L = 0, \pm 1$

$\Delta m = 0, \pm 1$

$\Delta S = 0$

$E = \frac{\hbar^2 k^2}{2m} \quad V(x) = \begin{cases} 0, & |x| > b \\ -V_0, & |x| < b \end{cases}$

$E > 0$



$-\frac{\hbar^2}{2\mu} \frac{d^2 \psi}{dx^2} + V(x) \psi = E \psi$

$\frac{d^2 \psi}{dx^2} = -\frac{2\mu(E - V(x))}{\hbar^2} \psi$

$|x| > b \Rightarrow V(x) = 0, \quad k = \sqrt{\frac{2\mu E}{\hbar^2}}$

$\psi_1(x) = e^{ikx} + A_1 e^{-ikx}$

$\psi_2(x) = A_2 e^{ikx}$

$|x| < b \Rightarrow V(x) = -V_0, \quad \lambda = \sqrt{\frac{2\mu(E + V_0)}{\hbar^2}}$

$\psi_2(x) = B_1 e^{i\lambda x} + B_2 e^{-i\lambda x}$

$\psi_1(b) = \psi_2(b)$

$\psi_2(b) = \psi_3(b), \quad \psi_2'(b) = \psi_3'(b)$

$B_1 e^{i\lambda b} + B_2 e^{-i\lambda b} = A_2 e^{ikb}$

$B_1 e^{i\lambda b} - B_2 e^{-i\lambda b} = \frac{k}{\lambda} A_2 e^{ikb}$

$\therefore \psi_1(b) = \psi_2(-b), \quad \psi_1'(b) = \psi_2'(-b)$

$e^{-ikb} + A_1 e^{ikb} = B_1 e^{-i\lambda b} + B_2 e^{i\lambda b}$

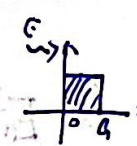
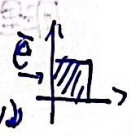
$e^{-ikb} - A_1 e^{ikb} = \frac{\lambda}{k} (B_1 e^{-i\lambda b} - B_2 e^{i\lambda b})$

解得 A_1, A_2, B_1, B_2

$T = \frac{4k\lambda'}{(k+\lambda')^2 \sin^2 \lambda' b + 4k^2 \lambda'^2}$ (透射系数)

$E > V_0 \Rightarrow \lambda = i\lambda', \quad \lambda' = \sqrt{\frac{2\mu(V_0 + E)}{\hbar^2}}$

$E > 0 > V_0 \Rightarrow \lambda = i\lambda', \quad \lambda' = \sqrt{\frac{2\mu(V_0 + E)}{\hbar^2}}$



$$\hat{H}' = \sum_{\alpha=1}^2 W_{\alpha} |e\rangle \langle \alpha| \quad |e\rangle = \frac{1}{\sqrt{2}} (|\alpha\rangle + |\beta\rangle)$$

$$\hat{H}'_{11} = \langle 1 | \sum_{\alpha=1}^2 W_{\alpha} |e\rangle \langle \alpha| | 1 \rangle$$

$$= C_1 W_1$$

$$\hat{H}'_{12} = \langle 1 | \sum_{\alpha=1}^2 W_{\alpha} |e\rangle \langle \alpha| | 2 \rangle$$

$$= C_1 W_2$$

$$\hat{H}'_{21} = C_2 W_1$$

$$\hat{H}'_{22} = C_2 W_2$$

$$\begin{pmatrix} C_1 W_1 & C_1 W_2 \\ C_2 W_1 & C_2 W_2 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\begin{pmatrix} C_1(C_1 W_1 + C_2 W_2) \\ C_2(C_1 W_1 + C_2 W_2) \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\begin{pmatrix} C_1 W_1 - E^{(0)} & C_1 W_2 \\ C_2 W_1 & C_2 W_2 - E^{(0)} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = 0$$

$$\therefore E_{\pm}^{(1)} = E_1 - E^{(0)} \quad E_{\pm}^{(2)} = E_1 - E^{(0)}$$

$$C_1^2 (W_1 + W_2) - C_2 E^{(0)} - W_2 = 0$$

$$C = \frac{E^{(0)} \pm \sqrt{E^{(0)2} + 4 W_2 (W_1 + W_2)}}{2(W_1 + W_2)}$$

四. (1) 基矢归一化

$$\psi_{10}(z, \vec{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a}\right)^{\frac{1}{2}} e^{-\frac{z}{a}} \quad \psi_{20}(z, \vec{r}) = \frac{1}{\sqrt{\pi}} \left(\frac{z}{a}\right)^{\frac{3}{2}} \left(1 - \frac{z}{2a}\right) e^{-\frac{z}{2a}}$$

$$\psi_{10}(z, \vec{r}, 0) = \sum_{nlm} C_{nlm} \psi_{nlm}(z, \vec{r}, 0)$$

$$\psi(z, \vec{r}, t) = \sum_{nlm} C_{nlm} \psi_{nlm}(z, \vec{r}, 0) e^{-i E_{nlm} t / \hbar}$$

$$|C_{nlm}|^2 = \int_0^{\infty} \psi_{nlm}(z, \vec{r}) \cdot \psi_{nlm}(z, \vec{r}) dz$$

$$P = |C_{100}|^2$$

Q: $|\psi(\omega)\rangle = |\psi\rangle e^{-i\omega t} + |\psi\rangle e^{-i\omega t}$

A: $P_{\psi} = |\langle \psi | \psi(\omega) \rangle|^2$

$= 1$, 总是1

五. $\hat{H} = \mu B \hat{S}_z$, $t=0$, $\psi(0) = a \chi_0 + b \chi_1$

(1) $|z = \frac{1}{2} \hbar, P_1 = \frac{1}{4} \hbar^2$

$|z = 0, P_2 = \frac{1}{4} \hbar^2$

(2) $\hat{H} = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

$$\frac{\hbar}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix} = E \begin{pmatrix} C_1 \\ C_2 \\ C_3 \end{pmatrix}$$

$$\lambda_1 = \hbar \quad |\lambda_1\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 0 \quad |\lambda_2\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$$

$$\lambda_3 = -\hbar \quad |\lambda_3\rangle = \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$P_1 = |\langle \lambda_1 | \psi \rangle|^2$$

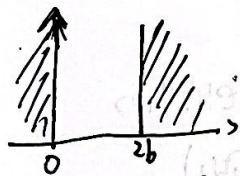
$$P_2 = |\langle \lambda_2 | \psi \rangle|^2$$

$$P_3 = |\langle \lambda_3 | \psi \rangle|^2$$



扫描全能王 创建

7.



(1) $\vec{F} = \vec{F}_1 + \vec{F}_2$

$$-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi_1(x) = E \psi_1(x)$$

$$-\frac{\hbar^2}{2\mu} \frac{d^2}{dx^2} \psi_2(x) = E \psi_2(x)$$

$$\psi_1(x) = \sqrt{\frac{1}{b}} \sin \frac{n_1 \pi}{2b} x, \quad E_1 = \frac{n_1^2 \pi^2 \hbar^2}{8\mu b^2}, \quad n_1 = 1, 2, 3, \dots$$

$$\psi_2(x) = \sqrt{\frac{1}{b}} \sin \frac{n_2 \pi}{2b} x, \quad E_2 = \frac{n_2^2 \pi^2 \hbar^2}{8\mu b^2}, \quad n_2 = 1, 2, 3, \dots$$

$$E = E_1 + E_2 = (n_1^2 + n_2^2) \frac{\pi^2 \hbar^2}{8\mu b^2}$$

(2) $\psi(x) = \frac{1}{b} \sin \frac{n_1 \pi}{2b} x \sin \frac{n_2 \pi}{2b} x$

$$E = \frac{\pi^2 \hbar^2}{4\mu b^2}$$

(3) $\vec{H} = \frac{e}{\mu c} \vec{B} \cdot \vec{S} = \frac{e \hbar}{2\mu c} \vec{B} \cdot \vec{\sigma} = \frac{e \hbar B}{2\mu c} \sigma_z$

$$(0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin up}}, (0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin down}}$$

$$(0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin up}}, (0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin down}}$$

$$(0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin up}}, (0, \frac{1}{2}, \frac{\hbar}{2})_{\text{spin down}}$$

1/2, 1/2

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \end{pmatrix} \otimes \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

